Schwartz Distribution

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Contents

1 Schwartz Distribution 1

I Test function and Schwartz Distribution 1

i Test function 1

ii Set of compact subsets as directed set 1

iii Topology of $C^k(K;U)$ and $C^k(U)$ 1

iv Canonical LF topology 1

v Distribution or Schwartz distribution 1

vi Support of distribution 1

II Induced distribution 2

i Distribution induced by measure 2

ii Distribution induced by L^1_{loc} function 2

III Extension and restriction 2

i Extension and restriction 2

ii Extension theorem 2

iii Vanish 3

iv Gluing theorem 3

IV Operation 3

i Multiplication by smooth function 3

ii Scaling operator 3

iii Translation operator 3

iv Derivative of distribution 3

v Convolution of functionals 3

vi Convolution of functional and distribution 4

vii Convolution of distributions 4

viii Cross-correlation 5
ix Fubini theorem 5
V Constant one distribution 5
VI Dirac delta function or unit impulse 5
i Fake function 5
ii Stieltjes integral 5
iii Dirac measure 5
iv Dirac distribution 6
v Convolution 6
vi Sinc function as nascent delta distribution 6
VII Heaviside (step) distribution or unit step distribution 6

1 Schwartz Distribution

I Test function and Schwartz Distribution

i Test function

A test function on U is a functional in $C_c^\infty(U)$. The space of test functions on U is $C_c^\infty(U)$, also denoted as $\mathcal{D}(U)$.

ii Set of compact subsets as directed set

The set of compact subsets of U form a directed set $\mathbb{K}$ with order by inclusion.

iii Topology of $C^k(K;U)$ and $C^k(U)$

For any subset U of $\mathbb{R}^n$, for any $k \in \mathbb{N} \cup \{\infty\}$, for any $f \in C^k(U)$, for any integer i such that $0 \leq i \leq k$, for any nonempty compact subset K of U, with the set $\mathcal{P}$ of all multi-indices p such that $|p| \leq i$, define,

$$q_{i,K}(f) \coloneqq \sup_{p\in\mathcal{P}} \left(\sup_{x\in K} |D^p f(x)| \right),$$

and define $q_{i,\varnothing}(f) = 0$.

Define the topology of $C^k(K;U)$ as the topology generated by the family of seminorms $q_{i,K}$ for any integer i such that $0 \leq i \leq k$, thus $C^k(K;U)$ is a locally convex topological vector space.

Define the topology of $C^k(U)$ as the topology generated by the family of seminorms $q_{i,K}$ for any compact subset K of U, for any integer i such that $0 \leq i \leq k$, thus $C^k(U)$ is a locally convex topological vector space.

iv Canonical LF topology

$C_c^k(U)$ is equipped with the canonical LF topology defined as the finest topology such that for any compact subset K of U, the inclusion map from $C^k(K;U)$ to $C_c^k(U)$ is continuous.

v Distribution or Schwartz distribution

A distribution or Schwartz distribution on U is a continuous linear functional on $C_c^\infty(U)$. The space of distributions on U is the strong dual space of $C_c^\infty(U)$, denoted as $\mathcal{D}'(U)$ and equipped with the strong dual topology.

vi Support of distribution

For any $T \in \mathcal{D}'(U)$, the support of T is defined as

$$\operatorname{supp}(T) = U \setminus \bigcup \{V \subseteq U \mid \rho_{VU}(T) = 0\}.$$

If $\operatorname{supp}(T)$ is a compact set, T is called with compact support. If $\operatorname{supp}(T)$ is a compact subset of a set $V \subseteq U$, T is called with compact support on V.

If $T \in \mathcal{D}'(U)$ is with compact support, for any $f \in C^\infty(U)$, $\langle T, f\rangle$ is defined as

$$\langle T, f\rangle \coloneqq \langle T, f\chi\rangle,$$

where $\chi \in \mathcal{D}(U)$ is an arbitrary function such that there exists a neighborhood N of $\operatorname{supp}(T)$ such that $\chi(N \cap U) = \{1\}$ and $f\chi \in \mathcal{D}(U)$. It's obvious that $\langle T, f\chi\rangle$ doesn't depend on the choice of χ.

II Induced distribution

i Distribution induced by measure

For a possibly signed Radon measure μ, the distribution $T_\mu \in \mathcal{D}'(U)$ induced by μ is defined by, for all $\varphi \in \mathcal{D}(U)$,

$$\langle T_\mu, \varphi\rangle = \int_U \varphi \,\mathrm{d}\mu\,.$$

ii Distribution induced by L^1_{loc} function

For a function $f \in L^1_{\text{loc}}(U)$, the distribution $T_f \in \mathcal{D}'(U)$ induced by f is defined by, for all $\varphi \in \mathcal{D}(U)$,

$$\langle T_f, \varphi\rangle = \int_U f(x)\varphi(x)\,\mathrm{d}x\,.$$

III Extension and restriction

i Extension and restriction

Let $V \subseteq U \subseteq \mathbb{R}^n$. For every function $f \in \mathcal{D}(V)$, the trivial extension of f to U is defined as a function $E_{VU}(f) \in \mathcal{D}(U)$ such that for all $v \in V$,

$$E_{VU}(f)(v) = f(v),$$

and for all $u \in U \setminus V$,

$$E_{VU}(f)(u) = 0.$$

This defines $E_{VU} : \mathcal{D}(V) \to \mathcal{D}(U)$ called trivial extension operator to U in V, which is a continuous injective linear map.

The transpose of E_{VU} is $\rho_{VU} : \mathcal{D}'(U) \to \mathcal{D}'(V)$ called restriction operator to V in U, and for all distribution $T \in \mathcal{D}'(U)$, $\rho_{VU}(T)$ is called restriction of T to V. A distribution in the image of ρ_{VU} is called extendable to U. A distribution in the image of $\rho_{V\mathbb{R}^n}$ is called extendable.

ii Extension theorem

The image of E_{VU} is a vector space. However, for all $V \subset U$, E_{VU} is not a topological embedding and that the subspace topology of the image of E_{VU} is strictly finer than the topology on the image of E_{VU} that makes E_{VU} a homeomorphism.

iii Vanish

$T \in \mathcal{D}'(U)$ is said to vanish in a subset V of U if for all $f \in \mathcal{D}(U)$, $\langle T, f \rangle = 0$.

iv Gluing theorem

Let $(U_i)_{i \in I}$ be an indexed family of open subsets of $\mathbb{R}^n$. For each $i \in I$, let $T_i \in \mathcal{D}'(U_i)$ and suppose that for all $i, j \in I$, the restriction of T_i to $U_i \cap U_j$ is equal to the restriction of T_j to $U_i \cap U_j$. Then there exists unique $T \in \mathcal{D}'\left(\bigcup_{i \in I} U_i\right)$ such that for all $i \in I$, the restriction of T to U_i is equal to T_i.

IV Operation

i Multiplication by smooth function

For any open subset $U \subseteq \mathbb{R}^n$, for any $T \in \mathcal{D}'(U)$, for any $f \in C^\infty(U)$, the distribution fT is defined by, for any $g \in \mathcal{D}(U)$,

$$\langle fT, g \rangle = \langle T, fg \rangle.$$

ii Scaling operator

For any open subset $U \subseteq \mathbb{R}^n$, for any $T \in \mathcal{D}'(U)$, T scaled by $a \in \mathbb{R}$, $S_a T$ or $T(a\cdot)$, is defined by, for any $f \in \mathcal{D}(U)$,

$$\langle S_a T, f \rangle = \frac{1}{|a|^n} \langle T, f\left(\frac{\cdot}{a}\right) \rangle.$$

iii Translation operator

For any open subset $U \subseteq \mathbb{R}^n$, for any $T \in \mathcal{D}'(U)$, T translated by $a \in \mathbb{R}^n$, $\tau_a T$ or $T(\cdot - a)$, as a member of $\mathcal{D}'(\{u + a \mid u \in U\})$, is defined by, for any $f \in \mathcal{D}(\{u + a \mid u \in U\})$,

$$\langle \tau_a T, f \rangle = \langle T, \tau_{-a} f \rangle.$$

iv Derivative of distribution

For any open subset $U \subseteq \mathbb{R}^n$, for any $T \in \mathcal{D}'(U)$, for any $\varphi \in \mathcal{D}(U)$, for any multi-index $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)$, the distributional derivative $D^\alpha T$ of T is defined by,

$$\langle D^\alpha T, \varphi \rangle = (-1)^{|\alpha|} \langle T, D^\alpha \varphi \rangle.$$

v Convolution of functionals

For any functionals f, g defined on $\mathbb{R}^n$, the convolution of f and g, denoted as $f * g$, is a functional on $\mathbb{R}^n$ defined as

$$f * g(x) = \int_{\mathbb{R}^n} f(y) g(x - y) \, \mathrm{d}y \,.$$

If $\operatorname{supp}(f) \cup \operatorname{supp}(g) \subseteq U$, then

$$f * g(x) = \int_U f(y) g(x - y) \, \mathrm{d}y \,.$$

Commutativity:

$$f * g = g * f.$$

Associativity:

$$(f * g) * h = f * (g * h).$$

Translation:

$$\tau_a(f * g) = \tau_a f * g = f * \tau_a g.$$

Differentiation:

$$D^\alpha(f * g) = D^\alpha f * g = f * D^\alpha g.$$

vi Convolution of functional and distribution

For any function $f \in C^\infty(\mathbb{R}^n)$, for any $T \in \mathcal{D}'(\mathbb{R}^n)$, with the constraint that either f or T is with compact support, the convolution of f and T, which is an element of $C^\infty(\mathbb{R}^n)$, denoted as $f * T$, is defined as

$$f * T \coloneqq x \mapsto \langle T, f(x - \cdot)\rangle.$$

Translation:

$$\tau_a(f * T) = \tau_a f * T = f * \tau_a T.$$

Differentiation:

$$D^\alpha(f * T) = D^\alpha f * T = f * D^\alpha T.$$

vii Convolution of distributions

For any open subset $U \subseteq \mathbb{R}^n$, for any distributions $S, T \in \mathcal{D}'(\mathbb{R}^n)$ with the constraint that at least one of them is with compact support, for any $f \in \mathcal{D}(\mathbb{R}^n)$, the convolution of S and T, which is an element of $\mathcal{D}'(U)$, denoted as $S * T$, is defined as

$$\langle S * T, f\rangle \coloneqq \langle S, x \mapsto \langle T, \tau_{-x} f\rangle\rangle.$$

Commutativity:

$$S * T = T * S.$$

Associativity:

$$(R * S) * T = R * (S * T).$$

Translation:

$$\tau_a(S * T) = \tau_a S * T = S * \tau_a T.$$

Differentiation:

$$D^\alpha(S * T) = D^\alpha S * T = S * D^\alpha T.$$

viii Cross-correlation

For functionals, functional and distribution, or distributions, f, g,

$$f \star g \coloneqq (\hat{R} f)^* * g.$$

Conjugate symmetry:

$$f \star g = (\hat{R}(g \star f))^*.$$

Translation:

$$\tau_a(f * g) = \tau_{-a} f * g = f * \tau_a g.$$

Differentiation:

$$D^\alpha(f * g) = D^\alpha f * g = f * D^\alpha g.$$

ix Fubini theorem

TODO

V Constant one distribution

For all $\varphi \in \mathcal{D}(U)$,

$$\langle 1, \varphi \rangle = \int_U \varphi(x) \, \mathrm{d}x \, .$$

VI Dirac delta function or unit impulse

i Fake function

The Dirac delta function $\delta(x)$ is often denoted as

$$\delta(x) \begin{cases} \infty, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

With the derivative of unit step function $H'(x)$ defined as $\delta(x)$.

ii Stieltjes integral

For any function f defined on $U \subseteq \mathbb{R}$, for any indexed family with $(A_i)_{i \in I} \subseteq U$,

$$\int_U f(x) \sum_{i \in I} \delta(x - A_i) \, \mathrm{d}x = \int_U f(x) \, \mathrm{d} \sum_{i \in I} H(x - A_i) = \sum_{i \in I} f(A_i).$$

iii Dirac measure

Dirac (delta) measure at a is a Radon measure defined, for any set A, as

$$\delta_a(A) \coloneqq \begin{cases} 1, & a \in A \\ 0, & a \notin A \end{cases}.$$

For any function f defined on $U \subseteq \mathbb{R}^n$, for any indexed family with $(A_i)_{i \in I} \subseteq U$,

$$\int_U f(x) \sum_{i \in I} \delta_{A_i}(x) \, \mathrm{d}x = \int_U f(x) \, \mathrm{d} \sum_{i \in I} \delta_{A_i}(x) = \sum_{i \in I} f(A_i).$$

iv Dirac distribution

Dirac (delta) distribution is the distribution induced by Dirac measure, that is, for any set $U \subseteq \mathbb{R}^n$, for any $a \in U$, for any $\varphi \in \mathcal{D}(U)$,

$$\langle \delta_a, \varphi \rangle = \varphi(a).$$

v Convolution

For all $a \in U \subseteq \mathbb{R}^n$, for all $f \in \mathcal{D}(U)$, for all $T \in \mathcal{D}'(U)$,

$$f * \delta_0 = f,$$

$$T * \delta_0 = T,$$

$$f * \delta_a = \tau_a f,$$

$$T * \delta_a = \tau_a T.$$

vi Sinc function as nascent delta distribution

For any complex-valued or real-valued $\varphi \in L^1(\mathbb{R})$:

$$\lim_{a \to 0} \int_{-\infty}^{\infty} \left(\frac{\sin\left(\frac{\pi x}{a}\right)}{\pi x} \right) \varphi(y - x) \, \mathrm{d}x = \varphi(y).$$

Proof. TODO □

VII Heaviside (step) distribution or unit step distribution

Heaviside (step) distribution or unit step distribution at a, denoted as H_a or u_a, is defined, for all $U \subseteq \mathbb{R}$, for all $\varphi \in \mathcal{D}(U)$, as

$$\langle H_a, \varphi \rangle = \int_{U \cap \mathbb{R}_{\geq a}} \varphi(x) \, \mathrm{d}x \, .$$

The derivative of it is

$$H_a' = \delta_a.$$